

Score-constrained ensembles and population transport

Restart, 2026-10-01. Sections 1–5 are written and internally checked: the constrained action, compact-shape obstruction, positive non-Gaussian forward orbit and nonlinear-copy recoil/tail test are established within their stated models. The unfinished calculation is a dynamical two-sheet realization on (18), retaining actual velocities and all force/switch energy. Start with the proposed balance equations in the Claude handout, which are candidate calculations, then test unread labels. Physical constraint enforcement and positive action selection have not been derived.

Restricting the ordinary action of two classical phase sheets to the shared-score preparations gives the canonical Fisher field action exactly. The restriction is not preserved by free Newtonian transport. A separate population-transfer construction identifies its full transport-conjugate action; bridge phase alone omits reservoir costs. The resulting two-coordinate model satisfies continuity, but its shape family fails the full field equation, even after adding Gaussian width. An evolving Gaussian-polynomial construction instead gives an exact positive, non-Gaussian solution for every forward time. A nonlinear coordinate copy generates a shape outside every finite Hermite–Gaussian family, with an exactly retained recoil-energy cost.

These are variational identities, an explicit field construction and stated closure obstructions, developed by GPT-6.1 Sol and GPT-6 Astra on 2026-10-01. They do not derive a physical constraint mechanism, quantum recording, or positive action necessity. The supplied parameter is $\kappa \geq 0$; the field obstruction below concerns $\kappa > 0$. Proof status: written, internally checked.

1. What the classical sheet action supplies

Work on a connected position interval or the line. Require smooth positive densities, finite displayed energies, and vanishing boundary terms for the variations used below. A classical phase sheet has position density r and momentum W_q . Its canonical one-form and free Hamiltonian are $\int W \delta r dq$ and $\int r W_q^2 / (2m) dq$, respectively, with $m > 0$.

This one-form follows from the ordinary particle form. If material coordinates a carry fixed probability mass, $q = X(a)$, $r dq = da$, and $v = \delta X \circ X^{-1}$ is a virtual displacement, then $\delta r = -\partial_q(rv)$. Integration by parts gives

$$\int W_q(X(a)) \delta X(a) da = \int r W_q v dq = \int W \delta r dq. \quad (1)$$

Fixed total mass makes a spatially constant shift of W a gauge. This is a smooth variational identity; no infinite-dimensional well-posedness theorem is being assumed.

For two sheets of probability mass 1/2, introduce the constraint

$$r_\sigma = \rho/2, \quad W_\sigma = S + \kappa \sigma \log(\rho/\rho_{\text{ref}}), \quad \sigma \in \{+1, -1\}. \quad (2)$$

Here ρ is normalized, S has action units, and the positive constant ρ_{ref} only makes the logarithm dimensionless. Changing it adds sheetwise constants and changes no momentum. Formula (2) is the field version of the supplied positive classical preparation in the recording note, §8.22.

Proposition 1 (exact constrained action). Pulling the two-sheet one-form and energy back to (2) gives

$$\begin{aligned} \Theta &= \sum_\sigma \int W_\sigma \delta r_\sigma dq = \int S \delta \rho dq, \\ \mathcal{H} &= \sum_\sigma \int \frac{r_\sigma W_{\sigma,q}^2}{2m} dq = \int \frac{\rho S_q^2}{2m} dq + \frac{\kappa^2}{2m} I(\rho), \\ I(\rho) &= \int \frac{\rho_q^2}{\rho} dq. \end{aligned} \quad (3)$$

The first identity cancels the two score signs; the second expands the two squares. Thus variation of the *restricted* action $\int[\int S\rho_t dq - \mathcal{H}]dt$ yields

$$\rho_t = -\partial_q(\rho S_q/m), \quad S_t = -\frac{S_q^2}{2m} + \frac{2\kappa^2}{m} \frac{(\sqrt{\rho})_{qq}}{\sqrt{\rho}}. \quad (4)$$

No further canonical-field bracket is needed after adopting the constraint and restricted variation. The Fisher modification is established prior art [HallReginatto2002; reading: passages pp. 3–8, especially the classical action (4) and modified action/equations (17)–(18); source companion]. The calculation here identifies which correlated sheet restriction produces that action and tests its invariance.

Why this is not an invariant reduction. Unrestricted classical sheet continuity gives, on (2),

$$\partial_t(r_+ - r_-) = -\frac{\kappa}{m} \partial_q^2 \rho. \quad (5)$$

For $\kappa > 0$, no strictly positive normalized density on the whole line has $\rho_{qq} \equiv 0$: an affine function cannot have those properties. Consequently every such constrained preparation departs from equal sheet densities immediately somewhere. Restricted variation and ordinary free sheet transport are different dynamics. Additional constraint reactions or state changes are required, as the exact Gaussian switching test in §8.23 also demonstrates. In particular (4)'s current uses the mean momentum S_q ; it does not give both physical sheet velocities $W_{\sigma,q}/m$ without further dynamics.

On connected positive support the labeled sheet momenta determine S_q and hence S up to a common constant. At exact vacuum, separate constants on disconnected components become invisible to the ordinary physical phase-space measure. This construction cannot carry them into a vacuum limit merely by declaring them canonical data.

2. Population transport includes the reservoirs

Let h_L, h_R be nonnegative smooth normalized bumps with compact, ordered, disjoint supports. Between them fix a bridge $B = [q_L, q_R]$. Choose a smooth positive normalized density ρ_0 with finite Fisher information. Put

$$\begin{aligned} h &= h_R - h_L, & \rho_w &= \rho_0 + wh, \\ G(q) &= \int_{-\infty}^q (h_L - h_R)(y)dy, & M(w) &= \int \frac{G(q)^2}{\rho_w(q)} dq, \\ R_B &= \int_B \frac{dq}{\rho_0(q)}. \end{aligned} \quad (6)$$

Here w is probability mass transferred from left to right, $0 \leq G \leq 1$, $G = 1$ on B , and G vanishes outside the reservoirs and their connector. If K is their compact enclosing interval, $c = \min_K \rho_0 > 0$ and $H = \|h\|_\infty > 0$, then $|w| < c/(2H)$ ensures $\rho_w \geq c/2$ on K . Thus all terms in (6) are finite, with $M(w) \leq 2|K|/c$, and smooth in this w range.

Vanishing exterior flux and continuity uniquely require $j(q) = \dot{w} G(q)$, because $\partial_q j = -\partial_t \rho_w$. The corresponding total transport kinetic energy is $mM(w)\dot{w}^2/2$. Both M and R_B have length-squared units. Smooth reservoir transport contributes positively, so $M(w) > R_B$.

Proposition 2 (continuity-compatible canonical family). Define an action field, up to a common constant, by

$$S_q(q; w, P) = \frac{PG(q)}{M(w)\rho_w(q)}. \quad (7)$$

The variable P has action units; it is not a point-particle momentum. Then

$$\begin{aligned}\Theta &= \int S \delta \rho_w dq = P dw, \\ H(w, P) &= \frac{P^2}{2mM(w)} + \frac{\kappa^2}{2m} I(\rho_w), \\ \dot{w} &= \frac{P}{mM(w)}, \quad \dot{P} = \frac{P^2 M'(w)}{2mM(w)^2} - \frac{\kappa^2}{2m} \frac{dI(\rho_w)}{dw}.\end{aligned}\tag{8}$$

These local reduced Hamilton equations preserve H while w remains in the positive-density range and satisfy the ambient continuity equation exactly.

Proof. Since $h = -G_q$, integration by parts yields $\int Sh dq = \int S_q G dq = P$. The phase energy in (3) is $P^2/(2mM)$, proving the Hamiltonian and its equations. Their first equation gives $\rho_w S_q/m = \dot{w}G$, the required current. $\square$

In this family the actual bridge action difference is

$$\Phi_B = S(q_R) - S(q_L) = \frac{R_B}{M(w)} P.\tag{9}$$

Thus Φ_B generally differs from the population-conjugate action $P = \int Sh_R dq - \int Sh_L dq$. The latter is a reservoir-weighted phase-action difference. Replacing it by bridge endpoint phases omits the reservoir motion. In coordinates (w, Φ_B) the canonical two-form has coefficient $M(w)/R_B$, rather than one.

For a fixed bridge density the exact dual identities are

$$E_{\text{transport}, B} = \frac{mJ^2 R_B}{2}, \quad \inf E_{\text{phase}, B} = \frac{\Phi_B^2}{2mR_B}, \quad J = \frac{\Phi_B}{mR_B}.\tag{10}$$

The phase infimum is the bridge lemma, (131), written with S instead of $\hbar\theta$; it is attained in the stated weighted H^1 class. The last identity makes the two energies equal. A low-density connector makes a fixed phase difference cheap and a fixed population flux expensive. It selects no absolute action scale.

The costs add over arbitrary spatial cuts, because both integrals of $1/\rho$ and of G^2/ρ add. This is an exact spatial transport identity, not Newton's temporal cell action. In the positive-tail family of recording (133), R_B grows as ϵ^{-2} . If reservoir costs additionally obey $M - R_B \leq C$ uniformly, then $|P - \Phi_B| \leq |P|C/R_B$ and the bridge becomes asymptotically dominant. That extra bound must be checked; exact vacuum is not such a check.

3. Continuity does not close the full field equation

Equations (8) are exact for the restricted model. They do not imply that its fixed density family is invariant under (4)'s Hamilton–Jacobi equation. For a Gaussian base $\rho_A = N(0, A)$ and initial $w = P = 0$, the full equation gives

$$\partial_t S_q = \frac{\kappa^2 q}{mA^2}.\tag{11}$$

The family (7) instead permits only a multiple of G/ρ_A at that initial time. It vanishes outside a compact interval, whereas (11) does not. This is an immediate nonzero transverse term for $\kappa > 0$. It cannot be removed by a spatially constant action gauge.

Adding Gaussian width supplies the missing global quadratic phase at $w = 0$. The next calculation tests whether that repairs closure for nonzero population transfer.

Proposition 3 (a fixed compact transfer shape cannot close). Fix $m, \kappa, A > 0$ and a nonzero $h \in C_c^\infty(\mathbb{R})$ with $\int h = 0$. The density family $\rho_{A,w} = \rho_A + wh$, allowing width variation and sufficiently small positive and negative w , cannot be locally invariant under (4) while including zero-phase-gradient states for every such w , satisfying continuity with vanishing

exterior flux, and having twice differentiable parameter trajectories. The same obstruction persists with Gaussian mean motion allowed. The claim concerns an open family of preparations, not all finite-dimensional models.

Proof. Set $r = \rho_A$ and $G = -\int_{-\infty}^q h$, so G is smooth and compactly supported. The integrated density tangents for width and transfer are $G_A = qr/(2A)$ and G . At an initially zero phase gradient, continuity first gives $\dot{A} = \dot{w} = 0$. Differentiating it and using (4) requires

$$\begin{aligned}\rho_w F_w &= c_A(w)G_A + c_w(w)G, \\ F_w &= \frac{2\kappa^2}{m} \partial_q \frac{(\sqrt{\rho_w})_{qq}}{\sqrt{\rho_w}}, \quad c_A = m\ddot{A}, \quad c_w = m\ddot{w}.\end{aligned}\tag{12}$$

Outside the compact supports, $\rho_w = r$ and $F_w = \kappa^2 q/(mA^2)$; hence $c_A(w) = 2\kappa^2/(mA)$ for every small w . At $w = 0$ this already matches the full expression, giving $c_w(0) = 0$. Evaluation at any point with $G \neq 0$ makes $c_w(w)$ smooth, since the left side is smooth while $\rho_w > 0$.

Put $u = h/r$. Differentiating the square-root ratio gives $\delta[(\sqrt{\rho})_{qq}/\sqrt{\rho}] = (u'' - qu'/A)/2$. Retaining the variation of the density multiplier in (12), the necessary linearized identity is

$$u''' - \frac{q}{A}u'' - \frac{u'}{A} + \frac{q}{A^2}u = cg, \quad g = G/r, \quad c = mc'_w(0)/\kappa^2.\tag{13}$$

Since $G' = -h$, one has $u = qg/A - g'$. Substitution yields

$$\begin{aligned}-g'''' + \frac{2q}{A}g''' + \left(\frac{4}{A} - \frac{q^2}{A^2}\right)g'' - \frac{4q}{A^2}g' \\ + \left(\frac{q^2}{A^3} - \frac{1}{A^2}\right)g = cg.\end{aligned}\tag{14}$$

The function g is smooth and compactly supported because $r > 0$. All four initial data vanish at a point outside its support. Uniqueness for this regular fourth-order linear ODE forces $g \equiv 0$, hence $h = -G' \equiv 0$, a contradiction. Allowing the Gaussian mean adds a term proportional to r in (12); tail matching forces its coefficient to zero and leaves the same contradiction. $\square$

The zero-phase-gradient and continuity assumptions are essential to the initial tangency test. The theorem permits isolated states, shapes depending on parameters or time, and other finite-dimensional families. It specifically rules out treating a fixed compact transfer bump plus Gaussian width as an exact free Fisher evolution. Generated shape information cannot be hidden inside that projection.

4. An exact positive nonlinear shape evolution

The compact-shape obstruction permits evolving tails. The following construction solves (4) exactly within its supplied dynamics. It also identifies a limitation of the first mode test: the linearized mode is a width/phase tangent, although its finite-amplitude completion is not Gaussian.

For $\kappa > 0$, introduce the auxiliary algebraic variable

$$\psi = \sqrt{\rho} e^{iS/(2\kappa)}, \quad \psi_t = \frac{i\kappa}{m} \psi_{qq}.\tag{15}$$

The second equation is equivalent to (4) on positive density: its real part gives $\rho_t = -\partial_q(\rho S_q/m)$ and its imaginary part gives (4)'s second equation after differentiating the displayed product. It is a change of variables in the adopted field equations, not an independent physical propagation premise.

Fix $A > 0$ with length-squared units and put

$$\begin{aligned} u &= \frac{\kappa t}{mA}, & B &= A(1 + u^2), & s &= \arctan u, \quad z = q/\sqrt{B}, \\ \psi &= B^{-1/4} e^{iuq^2/(4B)} \chi(z, s), & i\chi_s &= \left(-\partial_z^2 + \frac{z^2}{4} \right) \chi. \end{aligned} \quad (16)$$

For completeness, $a = \dot{B}/(2B) = \kappa u/(mB)$ and $b = \dot{s} = \kappa/(mB)$ obey $\dot{a} + a^2 = b^2$. Differentiating the second line cancels the terms $-a\chi/2$ and $-az\chi_z$ on both sides of (15). The remaining quadratic phase coefficient is $bz^2/4$, giving the last equation in (16). Thus no propagator formula is assumed.

Let

$$\begin{aligned} \phi_0(z) &= (2\pi)^{-1/4} e^{-z^2/4}, & H_2(z) &= (z^2 - 1)/\sqrt{2}, \quad \phi_2 = H_2\phi_0, \\ \psi_n(q, t) &= B^{-1/4} e^{iuq^2/(4B)} e^{-i(n+1/2)s} \phi_n(z), & n &\in \{0, 2\}. \end{aligned} \quad (17)$$

Direct differentiation gives the oscillator eigenvalues $1/2$ and $5/2$ for ϕ_0 and ϕ_2 . Gaussian moments give unit norms and orthogonality. Hence both displayed functions solve (15).

Proposition 4 (positive forward completion). For $0 < \epsilon < \sqrt{2}$, the normalized combination $\psi_\epsilon = (\psi_0 + i\epsilon\psi_2)/\sqrt{1 + \epsilon^2}$ gives a smooth solution of (4) on the whole line for every $t \geq 0$:

$$\begin{aligned} \rho_\epsilon(q, t) &= \rho_G(q, t) \frac{1 + 2\epsilon \sin(2s)H_2(z) + \epsilon^2 H_2(z)^2}{1 + \epsilon^2}, & \rho_G &= N(0, B), \\ S_\epsilon(q, t) &= \kappa \left(\frac{uq^2}{2B} - s \right) + 2\kappa \arctan \frac{\epsilon \cos(2s)H_2(z)}{1 + \epsilon \sin(2s)H_2(z)}, \\ \frac{\rho_\epsilon}{\rho_G} &\geq \frac{(1 - \epsilon/\sqrt{2})^2}{1 + \epsilon^2} > 0. \end{aligned} \quad (18)$$

The lower bound is relative to the Gaussian envelope; it supplies no absolute density floor at infinity. All displayed energies and moments are finite. The density is non-Gaussian for every $\epsilon > 0$.

Proof. Factoring out ψ_0 leaves $F = 1 + \epsilon[\sin(2s) + i\cos(2s)]H_2$. For forward time, $0 \leq s < \pi/2$, so $\sin(2s) \geq 0$; $H_2 \geq -1/\sqrt{2}$ gives $\Re F \geq 1 - \epsilon/\sqrt{2} > 0$. Thus the arctangent in (18) is one globally smooth phase, and $|F|^2$ gives its density and bound. Orthogonality gives exact normalization. Gaussian decay times a polynomial, with this strictly positive denominator, justifies the derivatives, moments and integrations by parts. The nonconstant quartic factor cannot be absorbed into a different Gaussian width. Equations (15)–(17) prove the field equations. $\square$

The conserved energy and first two even moments are explicitly

$$\begin{aligned} E &= \frac{\kappa^2(1 + 5\epsilon^2)}{2mA(1 + \epsilon^2)}, & EQ &= 0, \\ EQ^2 &= B \frac{1 + 2\sqrt{2}\epsilon \sin(2s) + 5\epsilon^2}{1 + \epsilon^2}, \\ EQ^4 &= B^2 \frac{3 + 12\sqrt{2}\epsilon \sin(2s) + 39\epsilon^2}{1 + \epsilon^2}. \end{aligned} \quad (19)$$

Indeed (3)'s energy equals $2\kappa^2 \int |\psi_q|^2/m$. Integration by parts in (15) makes its time derivative zero. At time zero the imaginary coefficient cancels the derivative cross term, and $\int |\phi'_0|^2 = 1/4$, $\int |\phi'_2|^2 = 5/4$, giving E . For the moments use $E_0 z^2 = 1$, $E_0 z^2 H_2 = \sqrt{2}$, $E_0 z^2 H_2^2 = 5$, and $E_0 z^4 = 3$, $E_0 z^4 H_2 = 6\sqrt{2}$, $E_0 z^4 H_2^2 = 39$. Here E_0 integrates against ϕ_0^2 . The variance has second derivative $4E/m$, as required by the free field virial identity; the linear-in-time term records the initial position/current correlation.

At first order in ϵ the paired tangents are

$$\delta\rho = 2\rho_G \sin(2s)H_2, \quad \delta S = 2\kappa \cos(2s)H_2. \quad (20)$$

They are width and quadratic-phase tangents, not an independent linear population-transfer mode. The result is their exact non-Gaussian nonlinear completion. It carries evolving tails rather than a fixed compact transfer bump. Restricting this one solution to any time partition is exactly consistent; no general continuum existence theorem or separated-packet apparatus has been constructed.

Forward time matters: at $s = -\pi/4$ the factor is $1 - \epsilon H_2$ and has real zeros. Thus (18) does not define a globally positive real-field family under all backward evolution. For fixed t , $\kappa \downarrow 0$ gives a stationary non-Gaussian density, $S \rightarrow 0$, and $E \rightarrow 0$. The construction leaves the zero branch admissible.

5. Recording generates a shape outside the finite mode family

Free completion must survive an actual intervention. Couple a body state (ρ, S) from (18), or any positive state with the following finite integrals, to a pointer configuration $Z \sim N(0, C)$, $C > 0$, with zero pointer action. Prepare the joint momenta with the shared score sign (2), so the full phase-space law need not factor. Apply the canonical copy (137) in the recording note, with smooth length-valued $f(q)$. In the algebraic variables its pullback is exactly

$$\Psi'(q, z) = \psi(q)\eta_C(z - f(q)), \quad \eta_C(y) = (2\pi C)^{-1/4}e^{-y^2/(4C)}. \quad (21)$$

The complete pointer-coordinate record $R = r$ gives

$$\rho_r(q) = \frac{\rho(q)e^{-(r-f(q))^2/(2C)}}{\int \rho(y)e^{-(r-f(y))^2/(2C)}dy}, \quad S_r(q) = S(q), \quad (22)$$

up to a record-dependent constant action. These are the exact conditional fields; the sign stays fair and independent of coordinates.

Proposition 5 (generated shape and retained recoil). For $f(q) = q^2/\ell$ with fixed length $\ell > 0$, every posterior (22) from (18) lies outside every finite Hermite–Gaussian expansion, even allowing a different Gaussian mean, width and quadratic phase. The mean body-energy increase is nevertheless exact:

$$\begin{aligned} \Delta E_{\text{body}} &= \frac{\kappa^2}{2mC} E_\rho[f'(Q)^2] = \frac{2\kappa^2}{mC\ell^2} E_\rho Q^2, \\ E_R I(\rho_R) - I(\rho) &= \frac{1}{C} E_\rho[f'(Q)^2]. \end{aligned} \quad (23)$$

All these posterior densities are smooth and positive, with finite Fisher information and body energy. The finite-mode failure is a failure of shape closure, not a violation of the conditional covariance bound (138).

Proof. The logarithmic posterior tail obeys $\lim_{|q| \rightarrow \infty} q^{-4} \log \rho_r(q) = -1/(2C\ell^2)$. A nonzero finite Hermite expansion over any Gaussian envelope has polynomial-times-Gaussian density and the same limit is zero. This proves the stated closure failure, without ruling out other finite models. For the energy differentiate (21): $\partial_q \Psi' = \psi_q \eta_C - f' \psi \eta'_C$. Integration in z kills the cross term since $\int \eta_C \eta'_C = 0$, and $\int |\eta'_C|^2 = 1/(4C)$. Multiplication by $2\kappa^2/m$ gives (23)'s first line, which is also the actual canonical recoil energy under (2). For its second line, the conditional score is $\partial_q \log \rho + (r - f)f'/C$; conditional on q , $r - f$ has mean zero and variance C . Expanding its square proves the identity. Quartic decay proves the stated posterior integrability. $\square$

If the coordinate record becomes unread, the marginal position density and mean current return to the original ρ and $\rho S_q/m$. The actual body energy keeps the increment (23). Resetting

its momenta to that marginal density's two score sheets would erase precisely this energy. Here every conditional phase gradient is the same, so there is no additional current-variance excess; for general branches both excesses in recording (135) must be retained. The pointer energy is unchanged by this impulsive shear. The extra body energy therefore requires work from its actuator; a closed energy-supplying apparatus has not been derived. Full conjugate access remains the distinct terminal escape (140). None of these formulas selects $\kappa > 0$.

6. Consequence for STATE

The canonical Fisher structure follows from a specified restriction of ordinary classical sheet action; physical preservation of that restriction remains open. Population memory must carry its full transport-conjugate action, including reservoirs, before taking a weak-bridge limit. The fixed compact-transfer family fails even with Gaussian width. Equations (18)–(20) instead complete a width/phase tangent to an exact positive non-Gaussian forward orbit. A nonlinear record immediately generates a quartic exponential tail outside the finite mode family, with its unread energy fixed by (23). Next lemma: carry the full generated density/phase and retained record labels through a dynamical constraint realization, first on this non-Gaussian orbit. It must keep the actual velocities and force/switch energy exchange, rather than project back to finite modes or reset an unread marginal. Physical apparatus closure and the two-unread-width test remain obligations. Reject a projected residual counted as exact closure, bridge phase used as population momentum without reservoir costs, or an unexcluded zero branch presented as positive action necessity.